



Course Name: Project/Thesis
Course Code: EEE 4000

A Dual-Stage Hybrid Random Forest Framework for Forward and Inverse Ion Implantation in Si, SiC, and GaAs Substrate

Abdullah Shadek Fahim | Student ID: 191131

Supervised by: Jahedul Islam, Lecturer, Dept. of EEE, JUST

Dept. of Electrical & Electronic Engineering
Jashore University of Science and Technology, Jashore-7408, Bangladesh

Date of Defense: March 07, 2026

Presentation Outline

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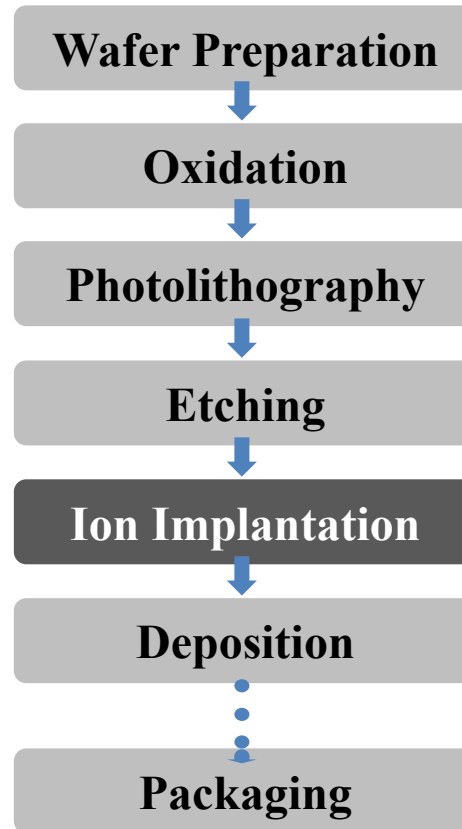


Introduction

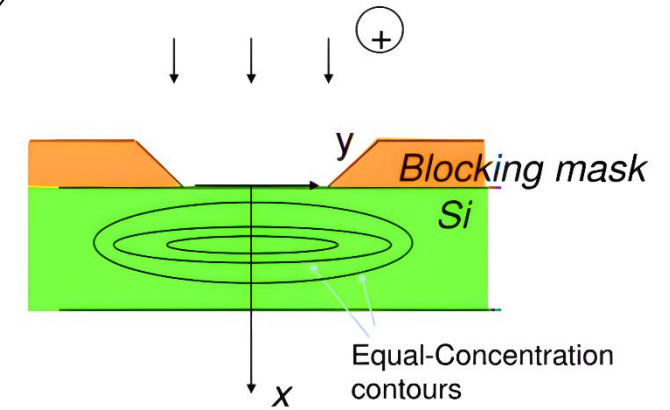
Packaged MOSFET [1]



Fabrication Process of MOSFET [2]



Ion Implantation [3]



Atomic Level After Implantation

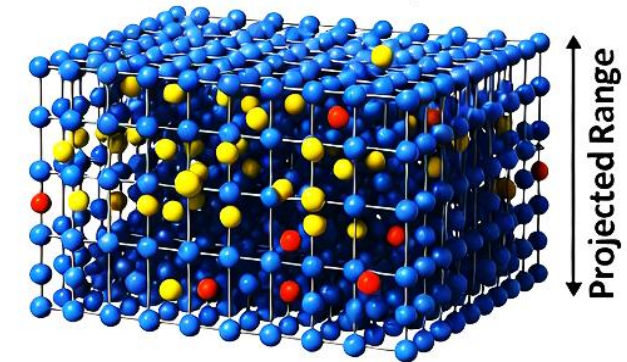


Fig. 1: Fabrication process of Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) focusing ion implantation

Introduction (Cont.)

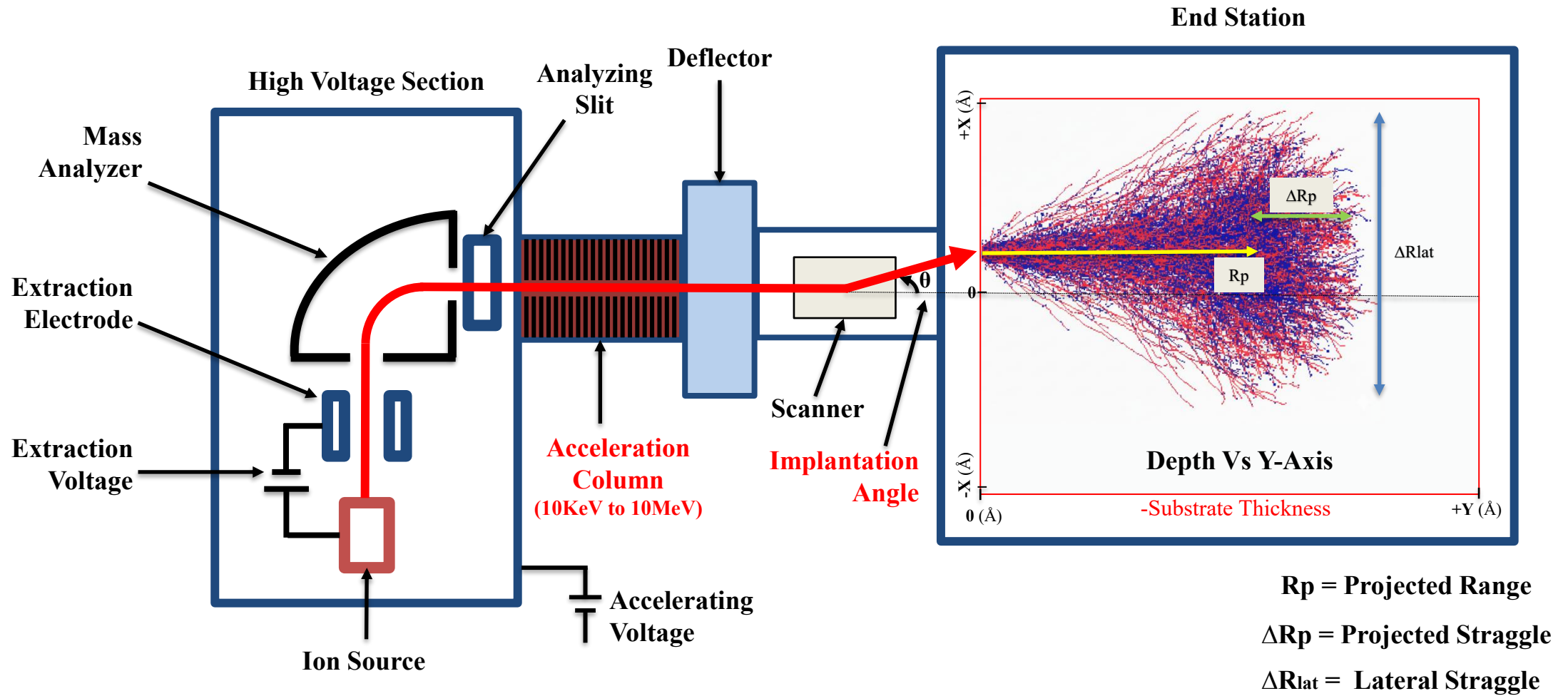


Fig. 2: Schematic experimental setup of ion implantation [4]

Introduction (Cont.)

FORWARD PROBLEM

Input Parameters $\rightarrow$ *Physical Outputs*

$$\mathbf{x} = \{ S, E, \theta, M, T \}$$

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) \dots\dots\dots(\text{eq. 1})$$

$$\mathbf{y} = \{ R_p, \Delta R_p, R_{lat}, \Delta R_{lat}, \\ R_{rad}, \Delta R_{rad}, V_{vac}, N_{back}, N_{trans} \}$$

$\rightarrow$ 9 physical outputs predicted simultaneously



INVERSE PROBLEM

Target Profile $\rightarrow$ *Implant Recipe*

$$\{ R_p^*, M, T \}$$

$$\mathbf{f}^{-1}(\mathbf{y}^*) = \mathbf{x} \dots\dots\dots(\text{eq. 2})$$

$$\mathbf{x} \rightarrow \{ (S_1, \theta_1, E_1), (S_2, \theta_2, E_2), \dots \}$$

$\rightarrow$ Multiple valid implant recipes returned

Problem Statements

Problem 1

The Simulation is Slow. The standard tool for simulating this process, *SRIM*, is accurate but time-consuming. We can't use it for rapid design iterations.

Problem 2

No Existing ML inverse model is established yet also the Inverse Problem is Hard. It's difficult to go backward from a desired material property to the specific implantation settings needed to achieve it. This is the main obstacle to rapid materials innovation.



Literature Review & Research Gap

Table 1: Comprehensive comparison of ion implantation and ML-based surrogate modeling approaches

Reference	Approach	Key Contribution	Limitation
LSS Theory (1963) [5]	Analytical	Gaussian approx. for range stats	Ignores channeling, compound materials
SRIM/TRIM (Ziegler 1980) [6]	Monte Carlo	Accurate ion–solid interaction	Forward-only, computationally expensive
Crystal-TRIM (Hobler 1996) [7]	Channeling	Crystalline effects in Si	Silicon-specific, not generalizable
SUPREM-IV / TCAD [8]	Analytical	Pearson-IV profiles for industry	No damage stats, limited physics range
Wang et al. (2024, DNN) [9]	Deep Learning	Inverse from optical PMOR signals	Black-box, Si-only, no depth profiles
Minagawa et al. (2024) [10]	ML	Range prediction with ML	Single output only, no inverse design
This Thesis (RF)	Ensemble ML	Physics-informed dual-stage RF	Based on SRIM simulated data only

Research Gap: No existing ML framework addresses both forward & inverse ion implantation across multiple substrates with physics awareness.



Thesis Objectives

1. To **create a dataset** by simulating different ions on different substrates for different parameters on SRIM.
2. To **automate the data generation process** from SRIM (TRIM.exe) using Python scripts to create a massive, diverse dataset.
3. To develop a **Hybrid Machine Learning Architecture** (Classifier + Regressor) that accurately handles both stopping and transmission events.
4. To perform **the inverse prediction** that can predict the correct SRIM input recipe from a desired set of outcomes (e.g., projected range).
5. To **validate the model** against standard SRIM data and ensure physical consistency (e.g., Energy vs. Range trends).



Methodology

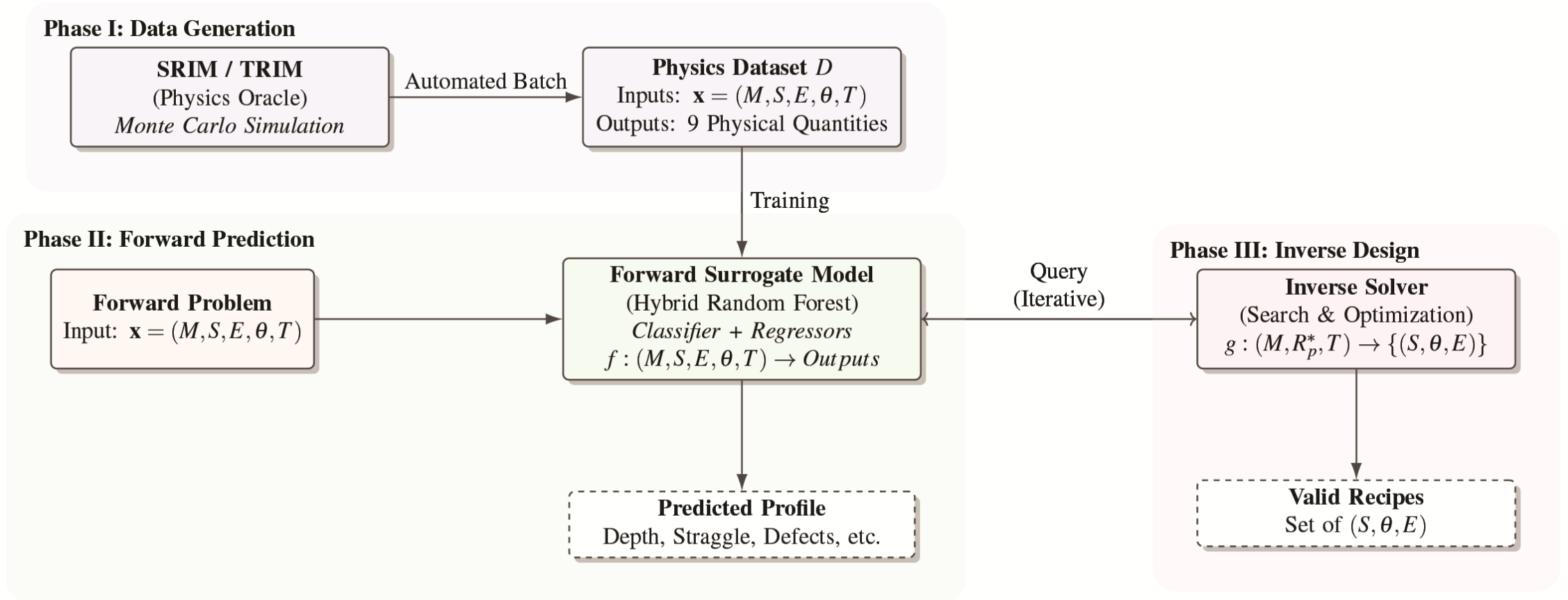


Fig. 3: Workflow of the dual-stage hybrid framework

Automated SRIM simulations generate a physics dataset. A hybrid Random Forest forward surrogate (classifier + regressors) emulates ion implantation behavior. An inverse solver queries the surrogate under physical constraints to generate a set of valid implantation recipes.

5,600

**Total SRIM
Simulations**

3

**Substrates
Si, SiC, GaAs**

5

**Ion Species
B, Mg, P, Ar, As**

9

**Output
Parameters**

Parameter Design Space

Energy, E 10 keV → 10 MeV

Angle, θ 0° → 89.9°

Thickness, T 100 Å → 10,000 Å

Ions, S B, Mg, P, Ar, As

Substrates, M Si, SiC, GaAs

9 Predicted Output Quantities

$R_p / \Delta R_p$ Projected range & straggle

$R_{lat} / \Delta R_{lat}$ Lateral range & straggle

$R_{rad} / \Delta R_{rad}$ Radial range & straggle

V_{vac} Vacancies per ion

N_{back} / N_{trans} Backscattered & transmitted ions

SRIM uses the Monte Carlo method within the Binary Collision Approximation (BCA) framework [11].

SRIM is a software suite used to calculate the interaction of ions with matter.

Developer James F. Ziegler
(IBM Research) 1980

Latest Version SRIM-2013

Layer Single to multilayer

Substrate Amorphous

Output Files RANGE.txt,
LATERAL.txt,
VACANCY.txt,
TDATA.txt

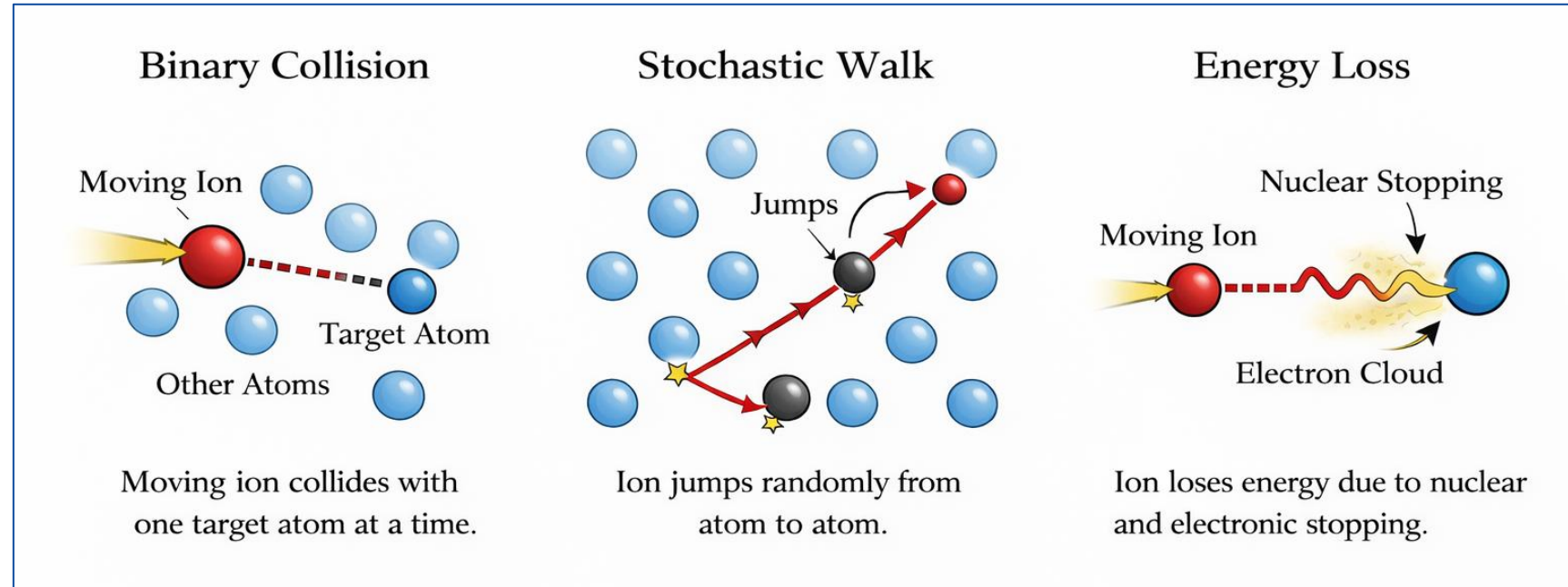


Fig. 4: The core mechanism of SRIM simulation [12]

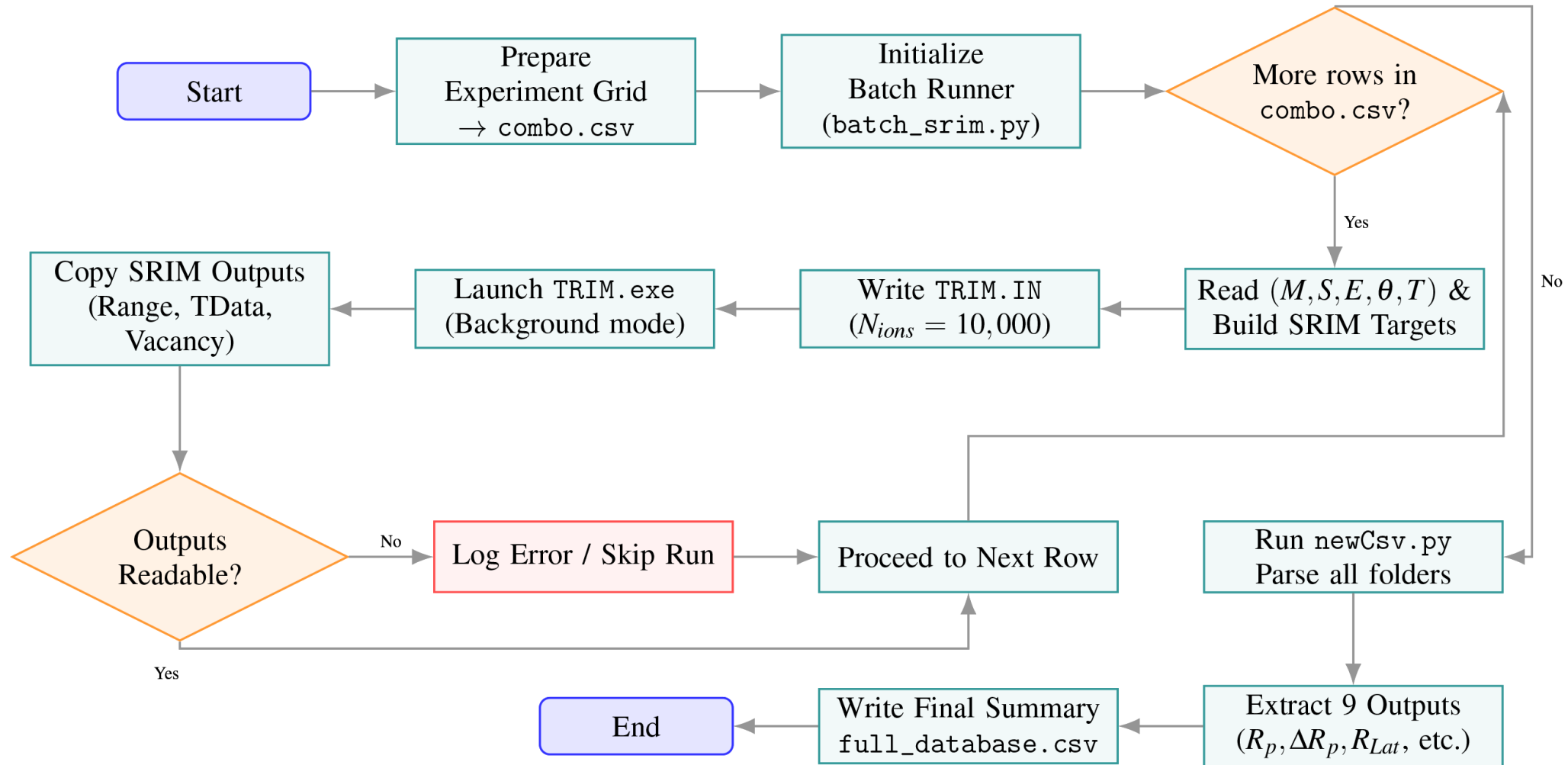


Fig. 5: The automated SRIM execution and data collection workflow

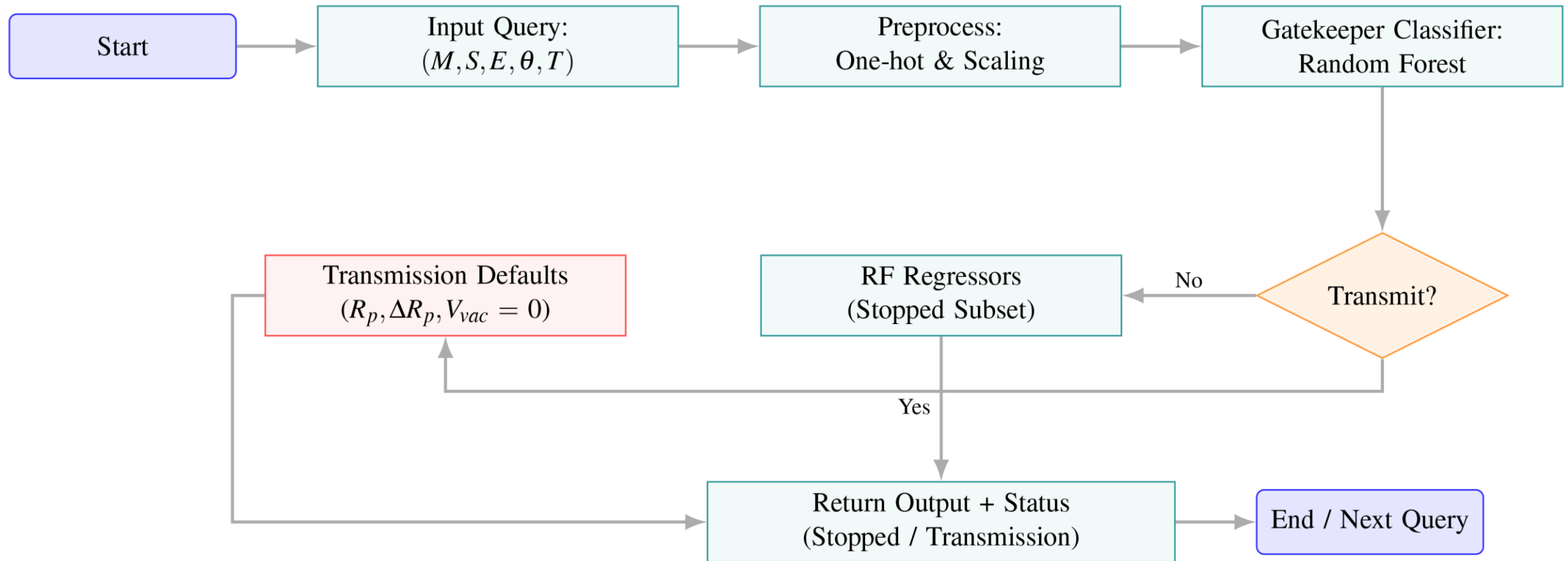


Fig. 6: The Proposed hybrid forward prediction workflow

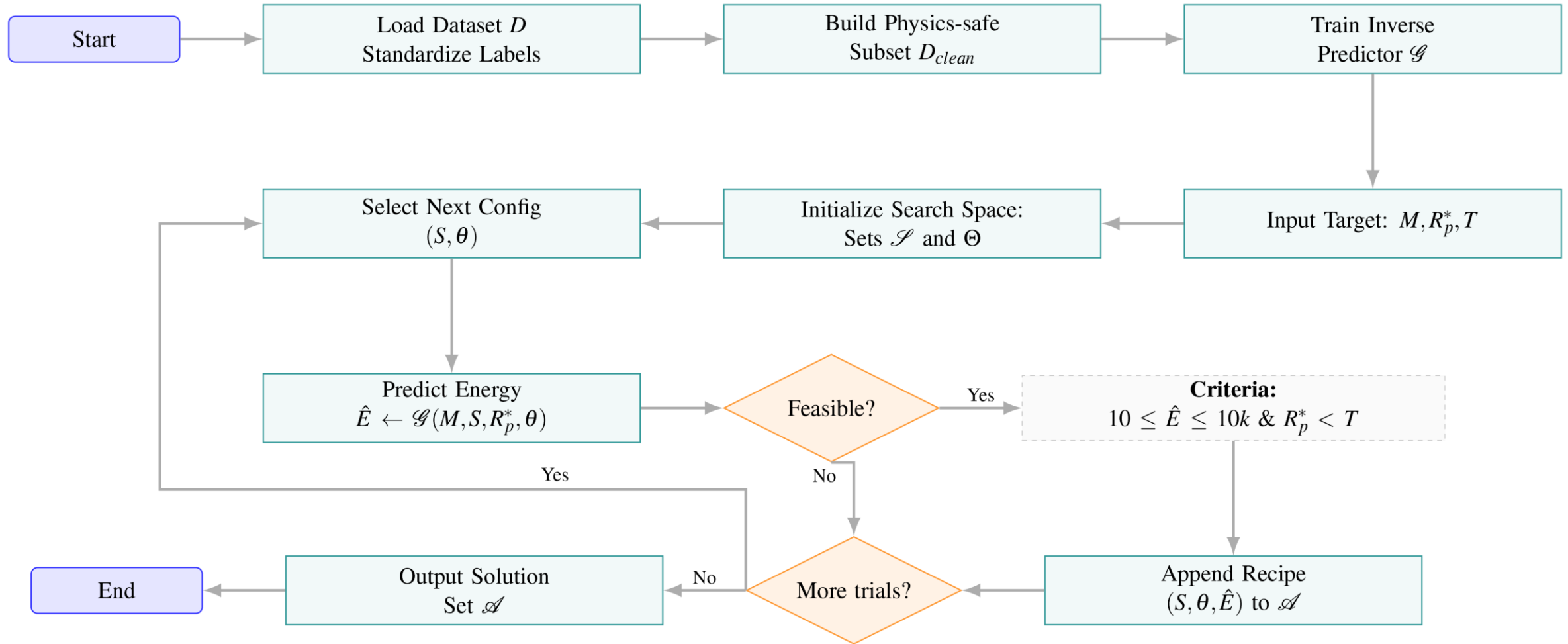


Fig. 7: Workflow of the physics-aware inverse solver system

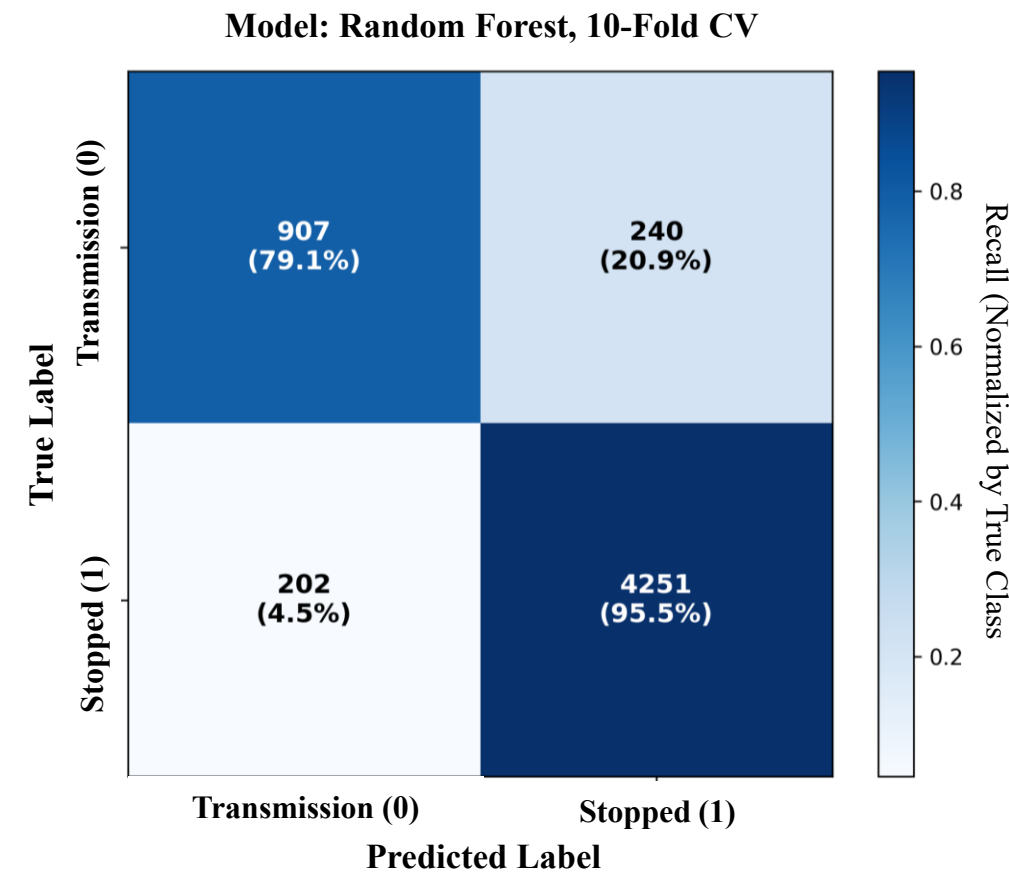


Fig. 8: Normalized confusion matrix for the Gatekeeper Random Forest classifier, highlighting the recall rates for Transmission (Class 0) and Stopped (Class 1) events.

Table 2: Comprehensive comparison of gatekeeper classification models

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.921	0.947	0.955	0.951
Gradient Boosting	0.908	0.935	0.942	0.938
KNN (k=3)	0.914	0.940	0.952	0.946
KNN (k=10)	0.913	0.944	0.948	0.946
KNN (k=5)	0.910	0.938	0.950	0.944



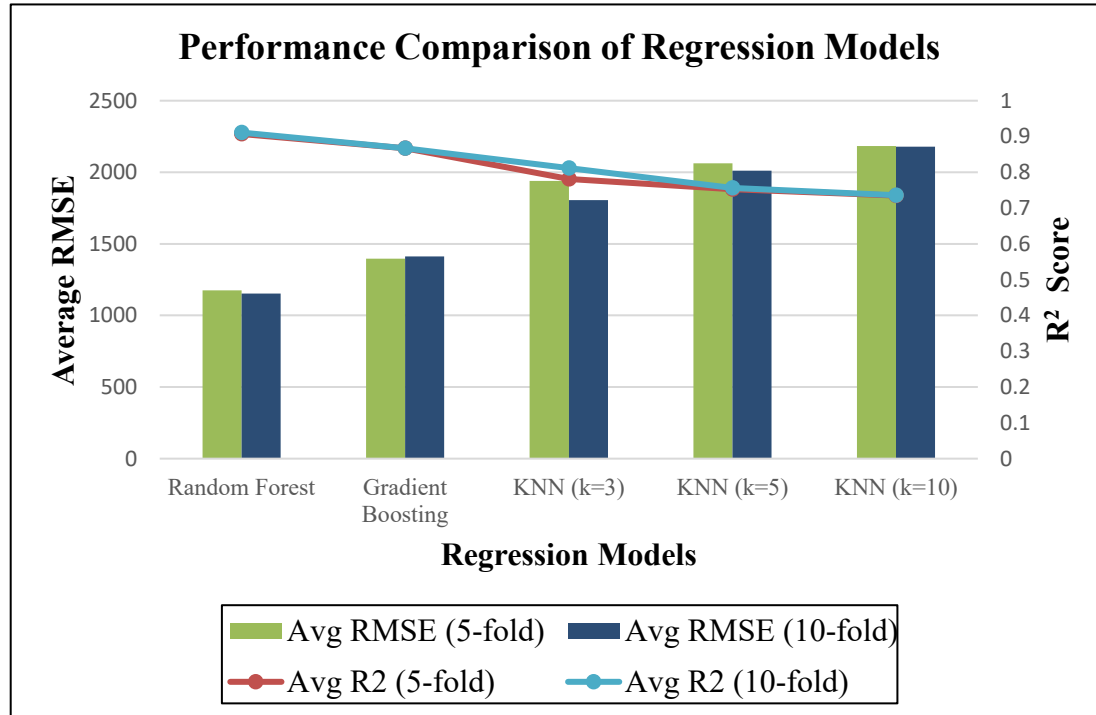


Fig. 9: Performance comparison of regression models (stopped ions only)

Table 4: Model performance across all nine target variables (Random Forest, 10-Fold CV)

Physical Output Target	MAE (Å / Count)	RMSE (Å / Count)	R ² Score
Projected Range (R_p)	233.03	515.28	0.945
Radial Range (R_{rad})	710.28	1861.65	0.977
Lateral Range (R_{lat})	790.72	2195.56	0.966
Vacancies per Ion (V_{vac})	536.56	1358.91	0.927
Backscattered Ions (N_{back})	52.41	165.71	0.985
Lateral Straggle (ΔR_{lat})	1096.73	3785.66	0.888
Longitudinal Straggle (ΔR_p)	111.85	278.30	0.811
Radial Straggle (ΔR_{rad})	369.49	940.53	0.659
Transmitted Ions (N_{trans})	514.86	1831.39	0.852

Table 5: Substrate-wise inverse regression metrics (RF)

Substrate	MAE (keV)	RMSE (keV)	R^2
GaAs	573	22.632	0.94
Si	567	67.974	0.92
SiC	732	82.588	0.96

Table 6: Ion-specific inverse model performance (RF, 10-fold cv)

Ion	MAE (keV)	RMSE (keV)	R^2
Ar	39.485	188.184	0.97
As	52.494	269.550	0.95
B	71.760	257.641	0.96
Mg	94.763	403.581	0.88
P	53.672	283.057	0.96

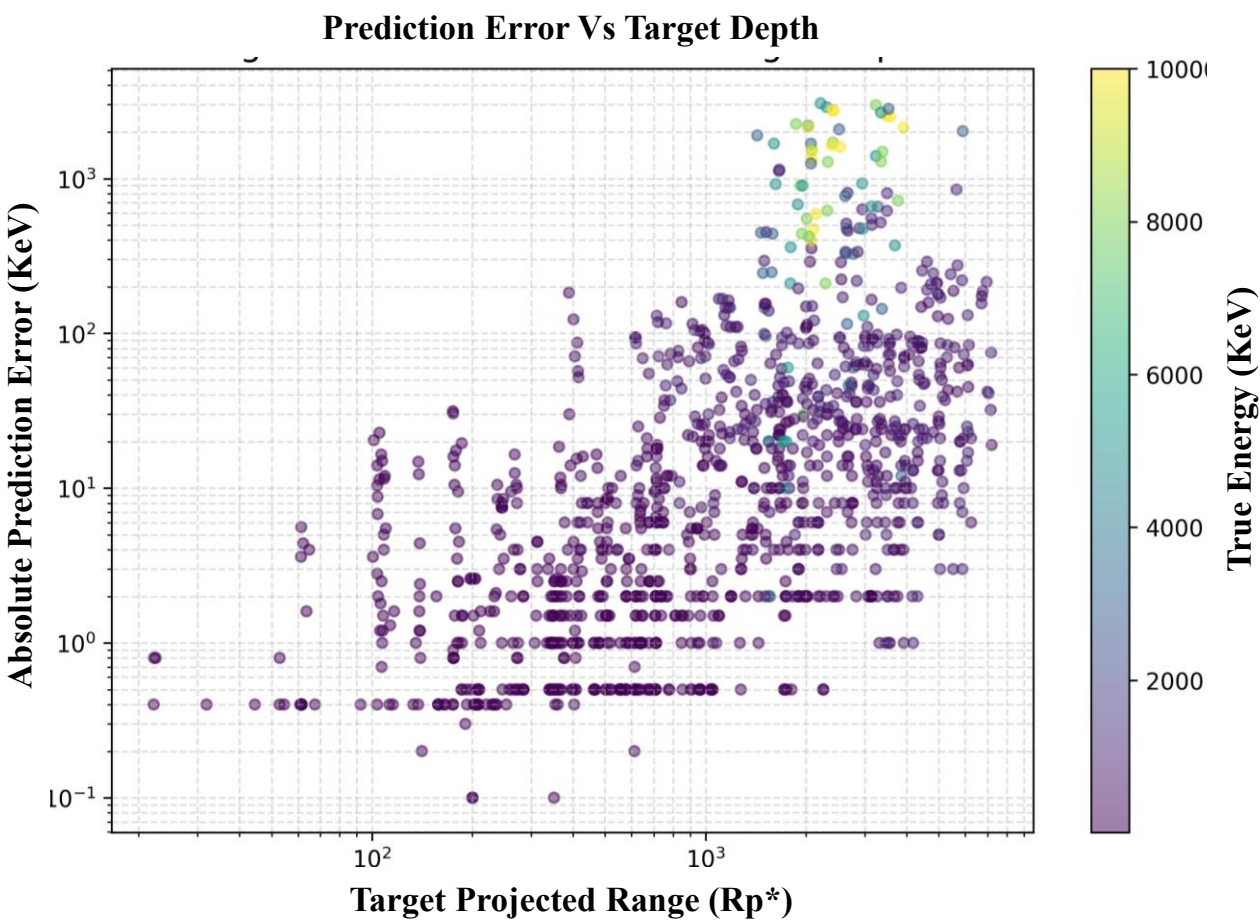


Fig. 10: Absolute inverse prediction error as a function of projected range to the target (Rp*).

186s

SRIM baseline
(10k ions, laptop)

0.45s

RF Forward Model
(constant time)

415×

Speedup
vs 10k SRIM Sim.

Table 7: Computational cost benchmarking with SRIM and RF model

Method	Ion Count	Hardware	Time/Run	Speedup
SRIM-2013	10,000	Core i5 Laptop	186.7 s	1× (Baseline)
SRIM-2013	10,000	Core i9 PC	67.7 s	~2.7×
SRIM-2013	100,000	Core i5 Laptop	2341 s	—
SRIM-2013	100,000	Core i9 PC	1141 s	—
RF Forward	—	Any hardware	0.45 s	415×
RF Inverse	—	Any hardware	0.35 s	533×

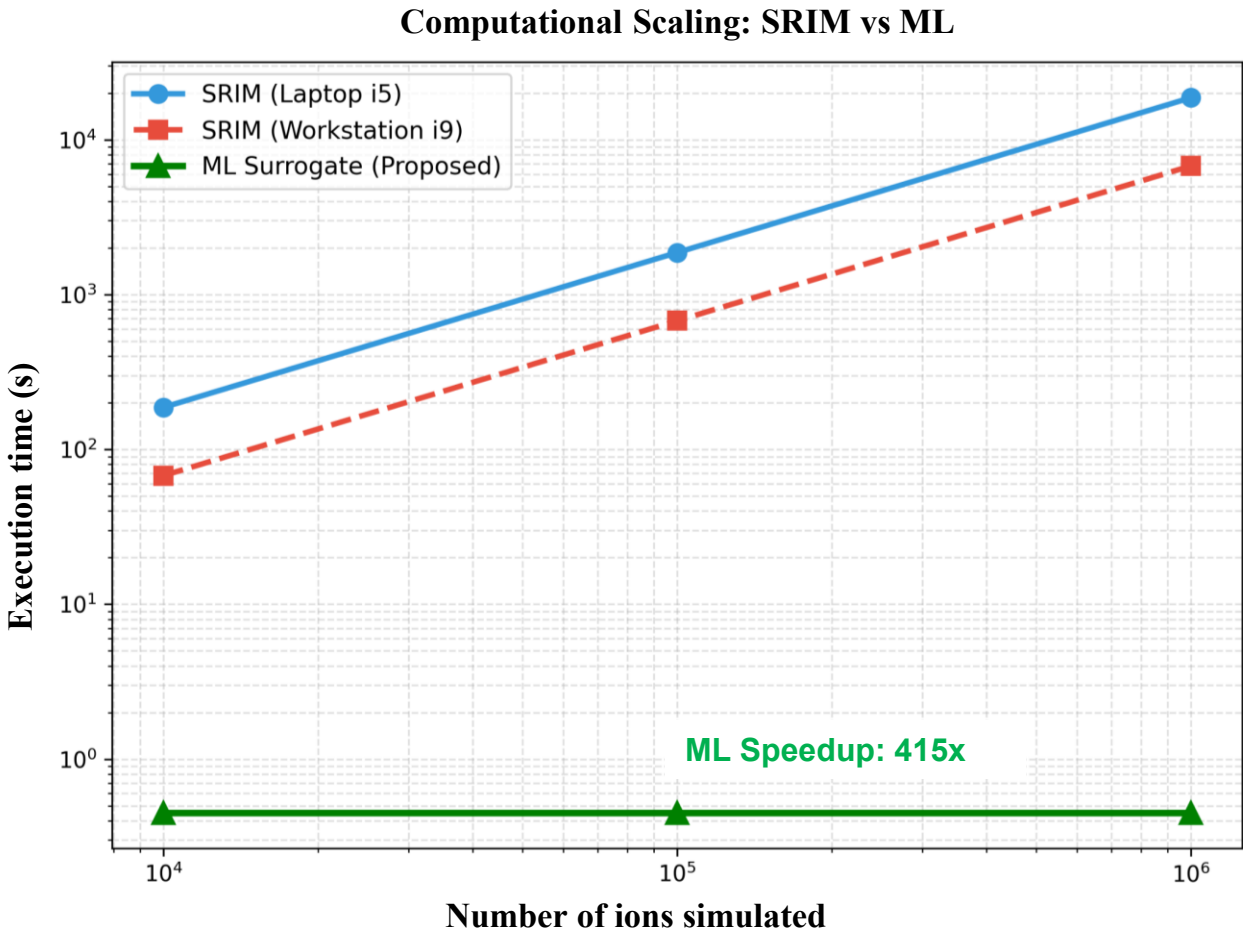


Fig. 11: Computational scaling analysis with SRIM simulation and proposed ML model

Conclusions — Key Contributions

A

Dual-Stage Forward–Inverse Framework

First ML system to jointly solve both forward prediction and inverse design for ion implantation that eliminating the need for manual SRIM iteration.

B

Physics-Informed RF Surrogate

Random Forest trained with domain-aware features achieves $R^2 > 0.98$ for projected range provides SRIM-level accuracy with millisecond inference.

D

Orders-of-Magnitude Speedup

415× faster than SRIM depending on fidelity mode that enables real-time parametric studies and digital twin applications.

E

Reproducible Virtual Lab Pipeline

End-to-end automated pipeline from SRIM data generation to ML inference to inverse solver that is fully documented and reproducible.

C

Cross-Material Generalization

One unified model works across Si, SiC, and GaAs without retraining which is first in literature to demonstrate this breadth.



Limitations & Future Work

Current Limitations

- 1 Amorphous SRIM Assumption**
Model trained on SRIM data assumes amorphous target that cannot capture channeling effects in crystalline Si or GaAs wafers.
- 2 No Annealing Physics**
As-implanted profile only. Post-implant dopant activation and transient-enhanced diffusion are not modeled.
- 3 No Experimental Validation**
Validated against SRIM (gold standard), not physical SIMS measurements that lack of access to experimental database.
- 4 Dose Independence**
Linear cascade approximation in dynamic amorphization at high doses and sputtering yield changes are not captured

Future Research Directions

- **PINNs Integration**
Physics-Informed Neural Networks to directly encode stopping power equations in the loss function
- **TCAD Training Data**
Re-train with Synopsys Sentaurus data for crystalline channeling accuracy
- **Experimental Validation**
Compare predictions against physical SIMS profiles from fabricated wafers
- **Implement Transfer learning**
By training on one material using transfer learning, implement the model for other materials
- **Generative AI for Inverse**
Use conditional GANs/VAEs for probabilistic inverse design with uncertainty quantification



References

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- [13]





Thank You

Questions & Discussion

5,600

SRIM Runs

9

Outputs

5

Ions

3

Substrates

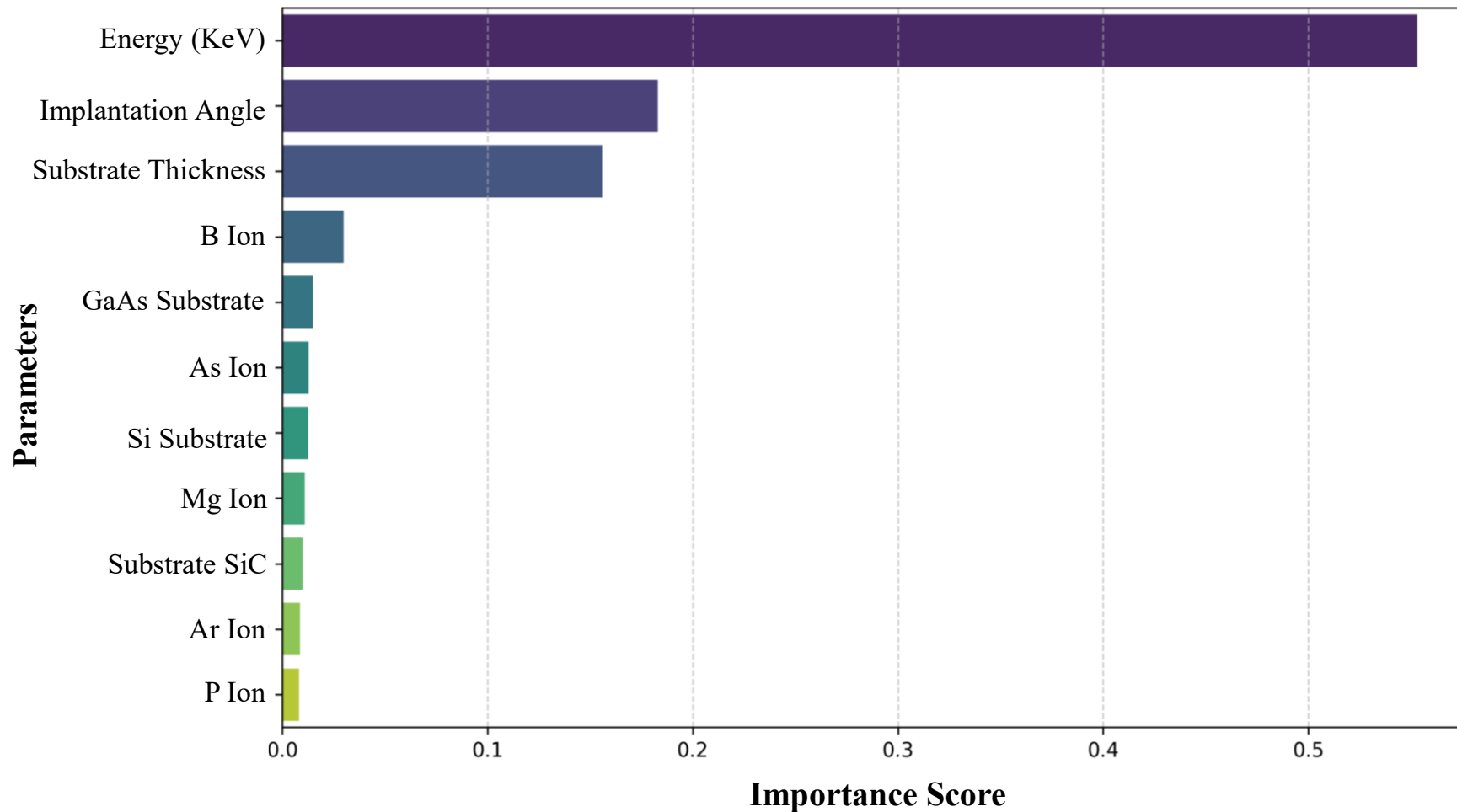
>0.98

Peak R^2

415×

Max Speedup

Data Analysis and Results



This is the Feature Importance plot for Classifier. It tells which parameters represent the decision-making process for detecting 'Zero-Range' events. **'Thickness, Angle and 'Energy** are the most dominant features. This is physically correct because whether an ion stops or transmits depends primarily on the substrate thickness and the ion's energy.

Fig. 12: Feature Importance for predicting Zero Range Events

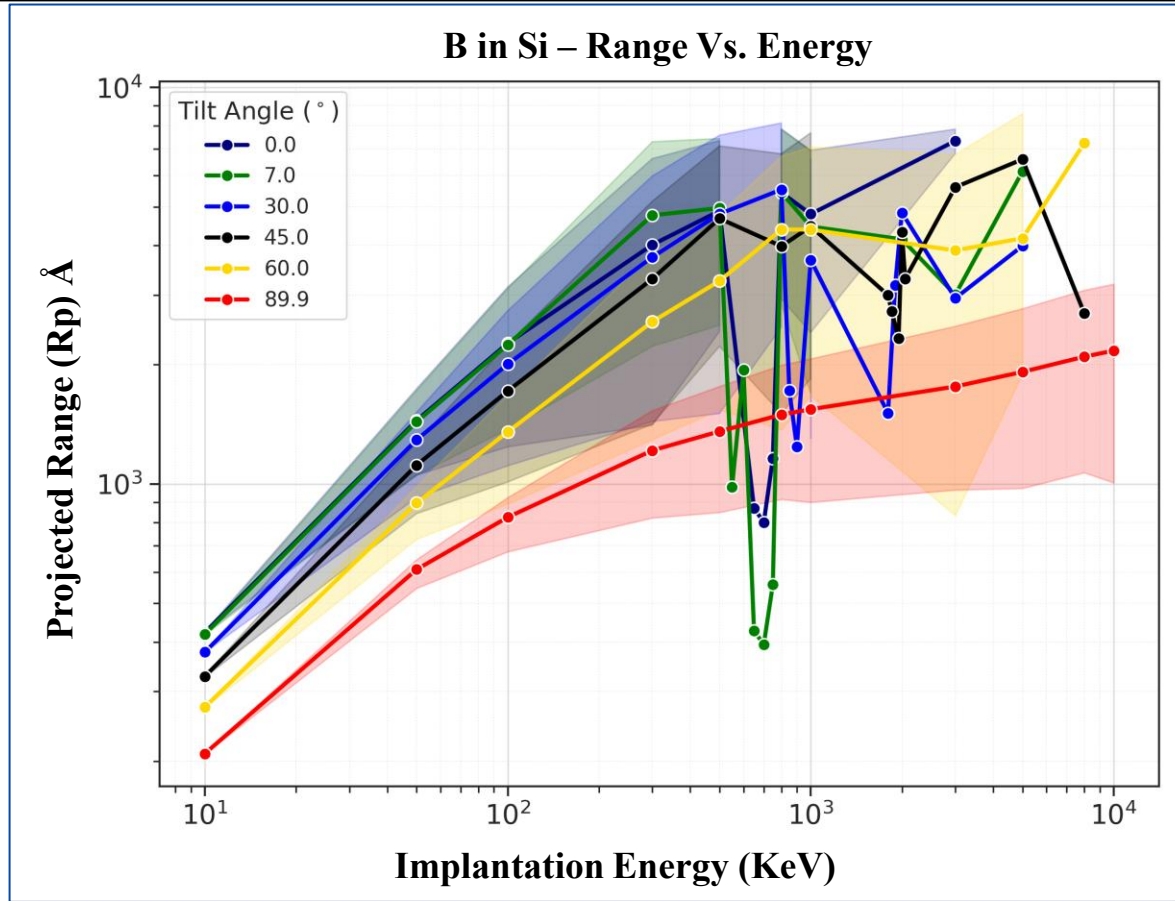


Fig. 13: Plot of R_p vs implantation energy for Boron ion into Silicon plotted for various tilt angles (0 to 89.9)

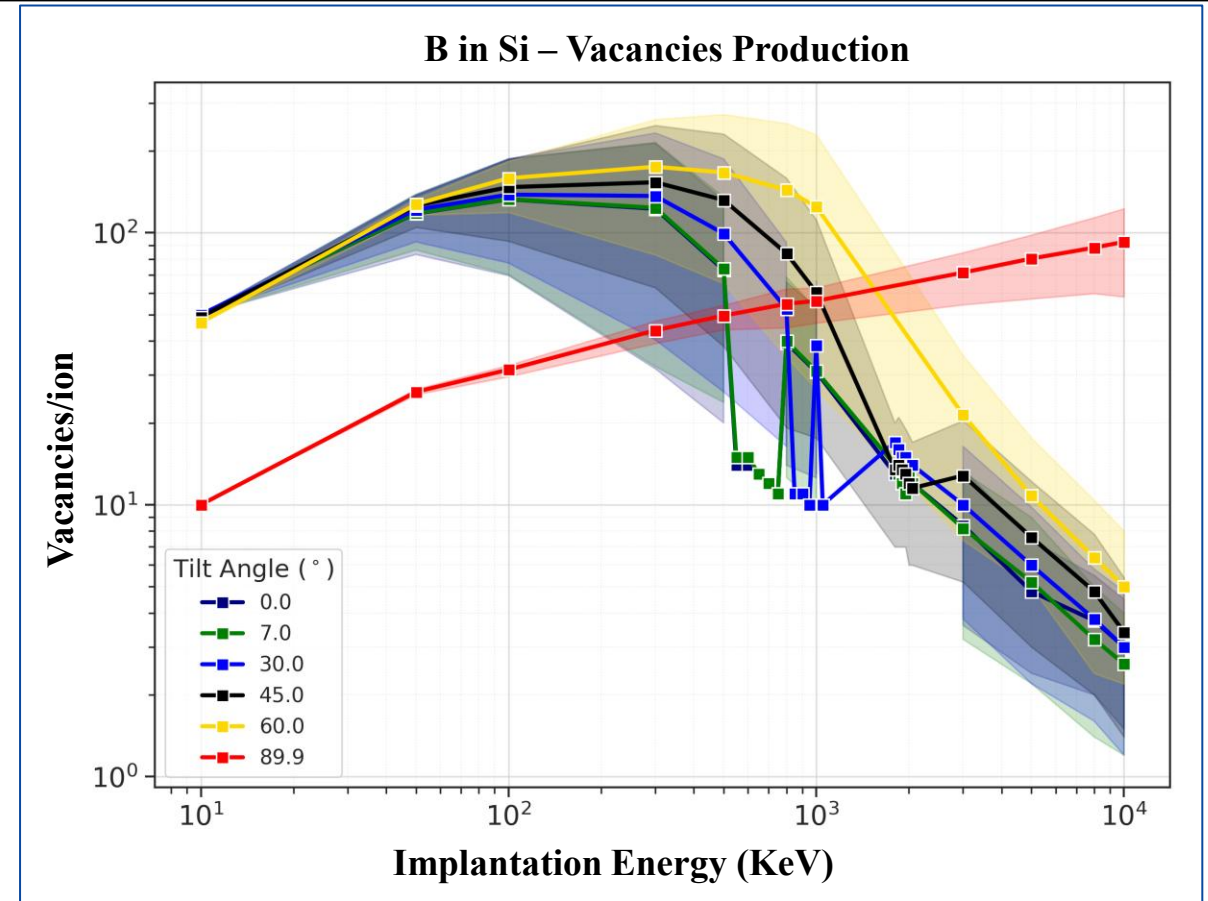


Fig. 14: Plot of vacancy production rate (Vacancies/Ion) with energy for Boron ion into Silicon Substrate.

LSS Theory & Gaussian Profile

LSS Theory (Lindhard–Scharff–Schjøtt, 1963)

- First analytical framework for ion range statistics
- Uses reduced energy ε and reduced distance ρ
- Predicts R_p , ΔR_p as functions of ion/target mass
- Assumes amorphous, homogeneous target material
- Ignores channeling effects in crystalline substrates

Table 8: Gaussian Model vs. Our RF Surrogate

Property	Gaussian (LSS)	RF (This Work)
Outputs	R_p , ΔR_p only	9 physical outputs
Multi-material	Separate tables	Single model
Damage (Vvac)	✗	✓ $R^2 = 0.927$
Backscatter	✗	✓ $R^2 = 0.985$
Inverse design	✗	✓ Native
Speed	Instant (formula)	0.45s (ML)

Gaussian Approximation

Dopant concentration: $C(z) = \frac{Q}{\sqrt{2\pi} \Delta R_p} \exp\left(-\frac{(z - R_p)^2}{2(\Delta R_p)^2}\right)$ (eq. 1)

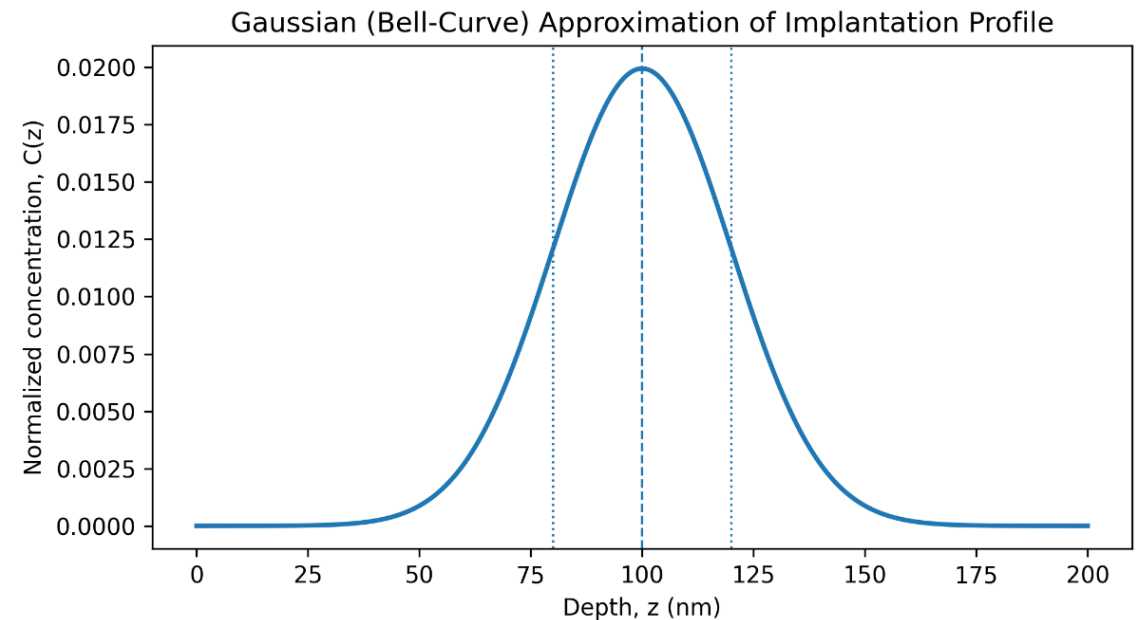


Fig. 15: Gaussian (bell-curve) approximation of the ion implantation depth profile showing the projected range R_p and longitudinal straggle

Generated using Python script for illustrative purposes.

Nuclear vs. Electronic Stopping Power

Electronic Stopping $S_e(E)$

- Inelastic collisions with electron cloud
- Dominant at HIGH energies (MeV range)
- Leads to deeper penetration, less damage
- Bethe–Bloch equation:

$$S_e(E) = \frac{4\pi z^2 e^4}{m_e v^2} n_e \left[\ln \left(\frac{2m_e v^2}{I} \right) - \ln(1 - \beta^2) - \beta^2 \right] \dots (eq. 2)$$

Nuclear Stopping $S_n(E)$

- Elastic collisions with lattice nuclei
- Dominant at LOW energies (keV range)
- Primary source of vacancy/damage production

$$E_{\max} = \frac{4M_1 M_2}{(M_1 + M_2)^2} E \dots \dots \dots (eq. 3)$$

$$\text{Stopping Power: } S(E) = S_n(E) + S_e(E) \dots \dots \dots (eq. 4)$$

$$\text{Projected Range: } R_p \approx \int_0^{E_0} \frac{dE}{S(E)} \dots \dots \dots (eq. 5)$$

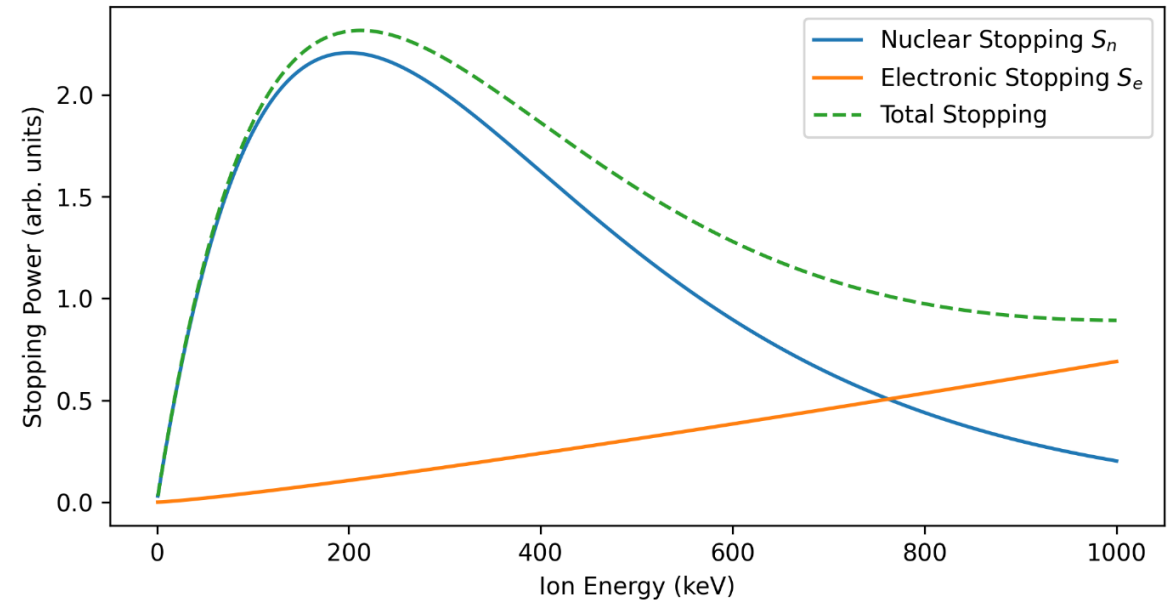


Fig. 16: Conceptual variation of nuclear and electronic stopping powers with ion energy.

Generated using Python script for illustrative purposes.

Substrate Crystal Structures & Properties

Silicon (Si)

- Diamond cubic structure
- **Bandgap: 1.12 eV**
- **Density: 2.33 g/cm³**
- Atomic number: Z = 14
- Displacement energy: ~15 eV
- **Industry standard CMOS substrate**

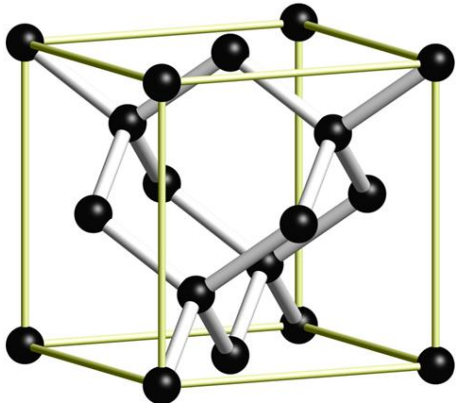


Fig. 17: Silicon diamond cubic unit cell

Silicon Carbide (4H-SiC)

- Hexagonal (4H polytype)
- **Bandgap: 3.26 eV (wide gap)**
- **Density: 3.21 g/cm³ (denser than Si)**
- Si: Z=14, C: Z=6 (binary lattice)
- Displacement energy: 20–35 eV (higher)
- **Power electronics applications (EV, 5G)**

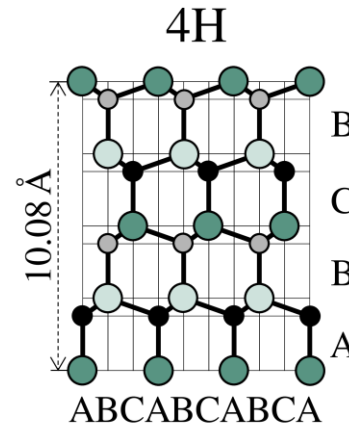


Fig. 18: 4H-SiC hexagonal polytype structure

Gallium Arsenide (GaAs)

- Zincblende crystal structure
- **Bandgap: 1.42 eV (direct)**
- **Density: 5.32 g/cm³ (heaviest)**
- Ga: Z=31, As: Z=33 (heavy nuclei)
- **RF/microwave & optoelectronic devices**

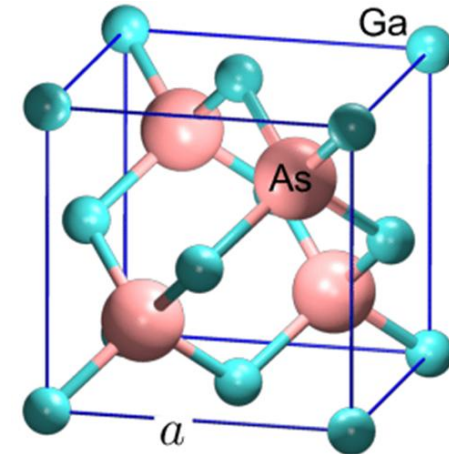


Fig. 19: GaAs zincblende unit cell.

<https://www.chemicalbook.com/article/diamond-crystal-structure-uses.htm>

Why Random Forest?

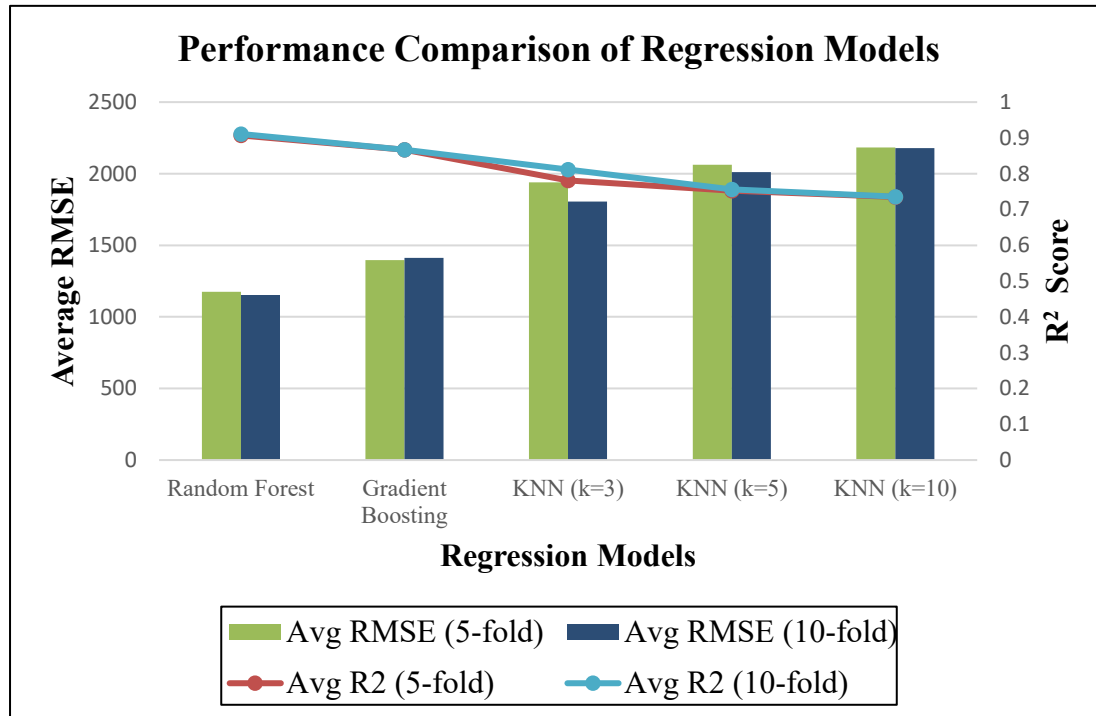


Fig. 20: Performance comparison of regression models (stopped ions only)

Why Random Forest Fits This Problem

- **Handles non-linear physics** without explicit equations
- **Mixed Data** - categorical + continuous inputs natively (substrate, ion, E, θ , T)
- **Piecewise physical modeling**: each tree covers one stopping regime
- Implicit regularization via **ensemble averaging** → **no overfitting**
- **Multi-output regression** through vector-valued leaves
- **Feature importance gives physical insight** (Energy > Angle > Thickness)
- Stable under small changes critical for inverse search loop
- **Interpretable**: can audit which features drive predictions

Physics-Aware Feature Encoding & Preprocessing

Table 9. Input of the model with different parameters

Parameter (Column Name)	Data Type	All Unique Values Used in Dataset
Substrate (3)	Categorical	Si, GaAs, SiC
Ion (5)	Categorical	Ar, As, B, Mg, P
Implantation Energy (11)	Numerical	10.0, 50.0, 100.0, 300.0, 500.0, 800.0, 1000.0, 3000.0, 5000.0, 8000.0, 10000.0
Implantation Angle (6)	Numerical	0°, 7°, 30°, 45°, 60°, 89.9°
Substrate Thickness (5)	Numerical	1000Å, 2000Å, 5000Å, 8000Å, 10000Å

One-Hot Encoding (Categorical Variables)

- Substrate $M \in \{\text{Si, SiC, GaAs}\} \rightarrow 3$ binary columns
- Ion $S \in \{\text{B, Mg, P, Ar, As}\} \rightarrow 5$ binary columns
- **Why NOT label encoding (Si=1, SiC=2)?**
 - Implies **false** ordinal relationship between materials
 - One-hot: each substrate treated as independent physics regime

Data Training and Evaluation Pipeline

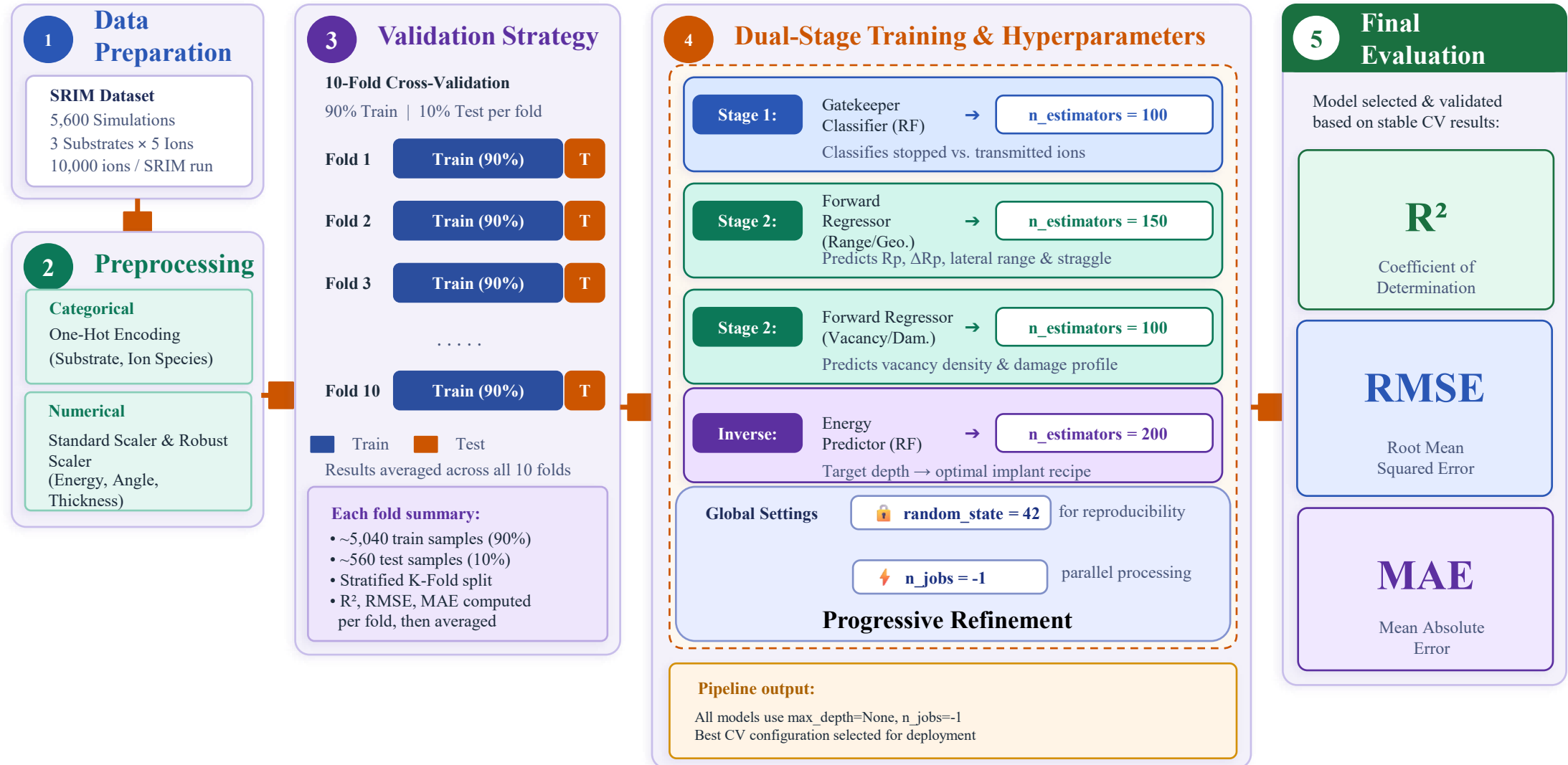
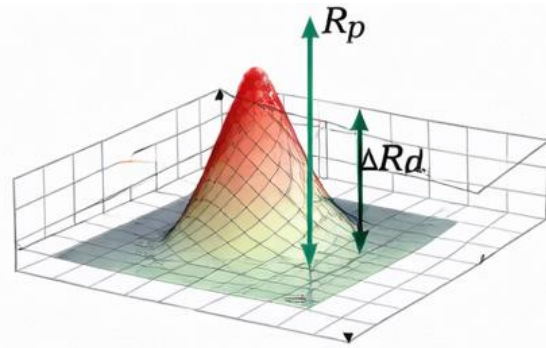
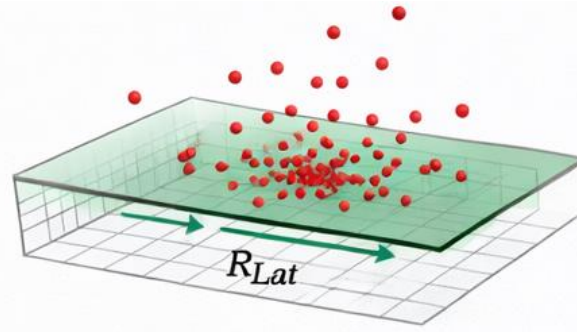


Fig. 21: Data training and evaluation pipeline

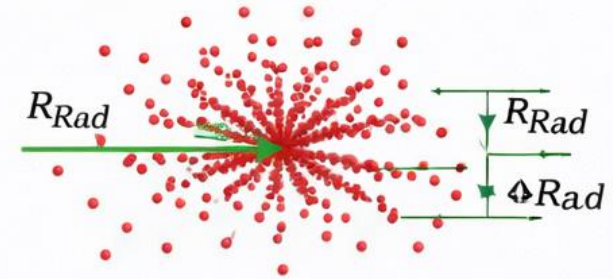
Predicted Output Parameters



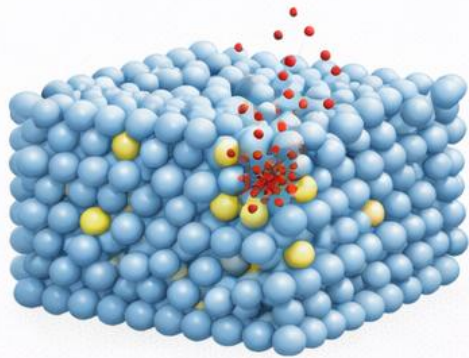
Projected Range (R_p and ΔR_p)



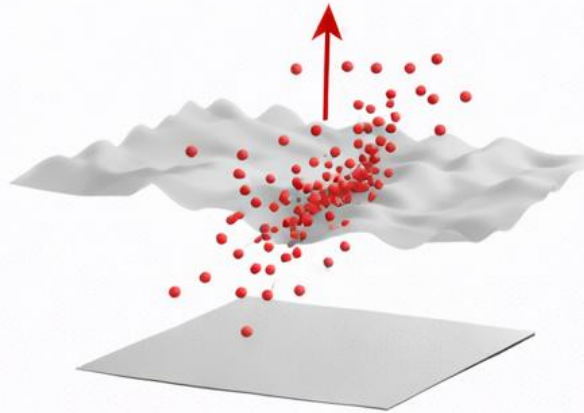
Lateral Range (R_{Lat} and ΔR_{Lat})



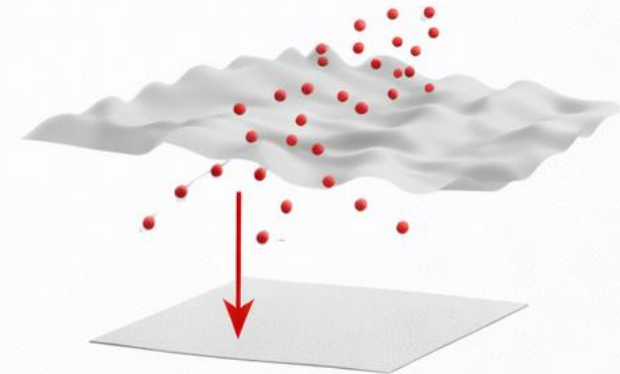
Radial Spread (R_{rad} and ΔR_{rad})



Vacancies (V_{vac})



Backscattered Ions (N_{back})



Transmitted Ions (N_{trans})

Fig. 22: Predicted output parameters of the model

Generated using LLM for illustrative purposes.

Experimental Validation

Validation vs. Janson et al. (2003) — SiC Target
B and P ions, angle = 0°, thickness = 10,000 Å

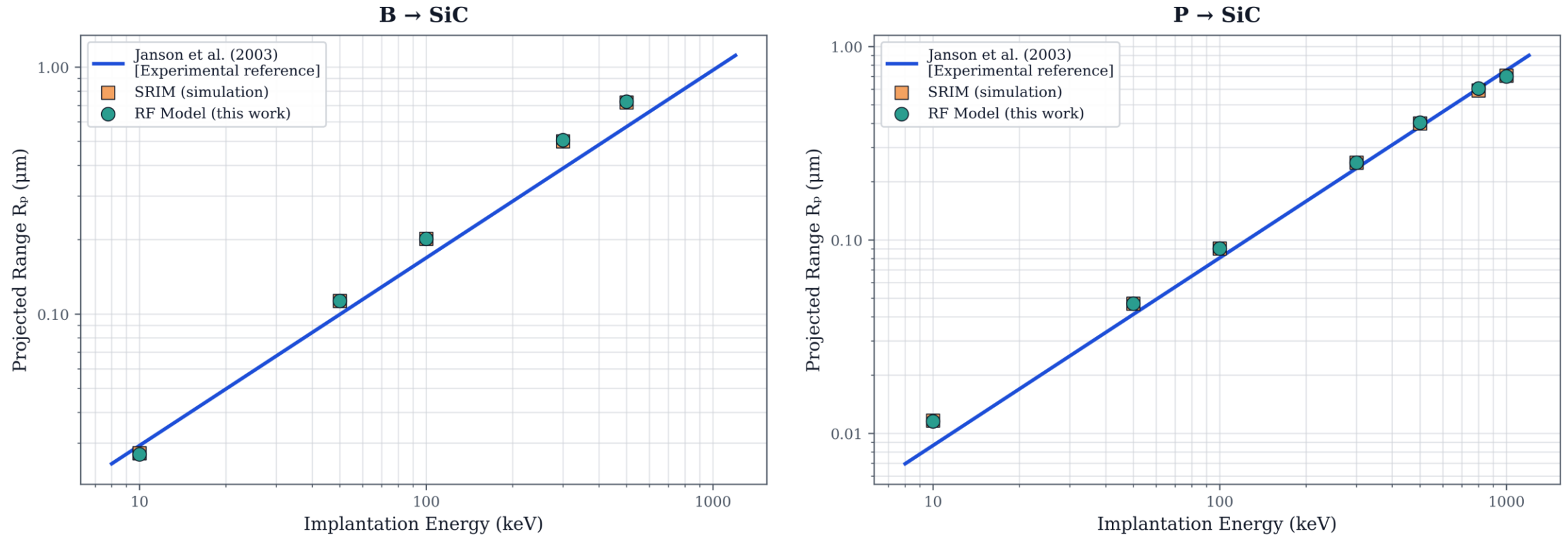


Fig. 23: Experimental SIMS validation with Model output for SiC substrate

Experimental Validation (Cont.)

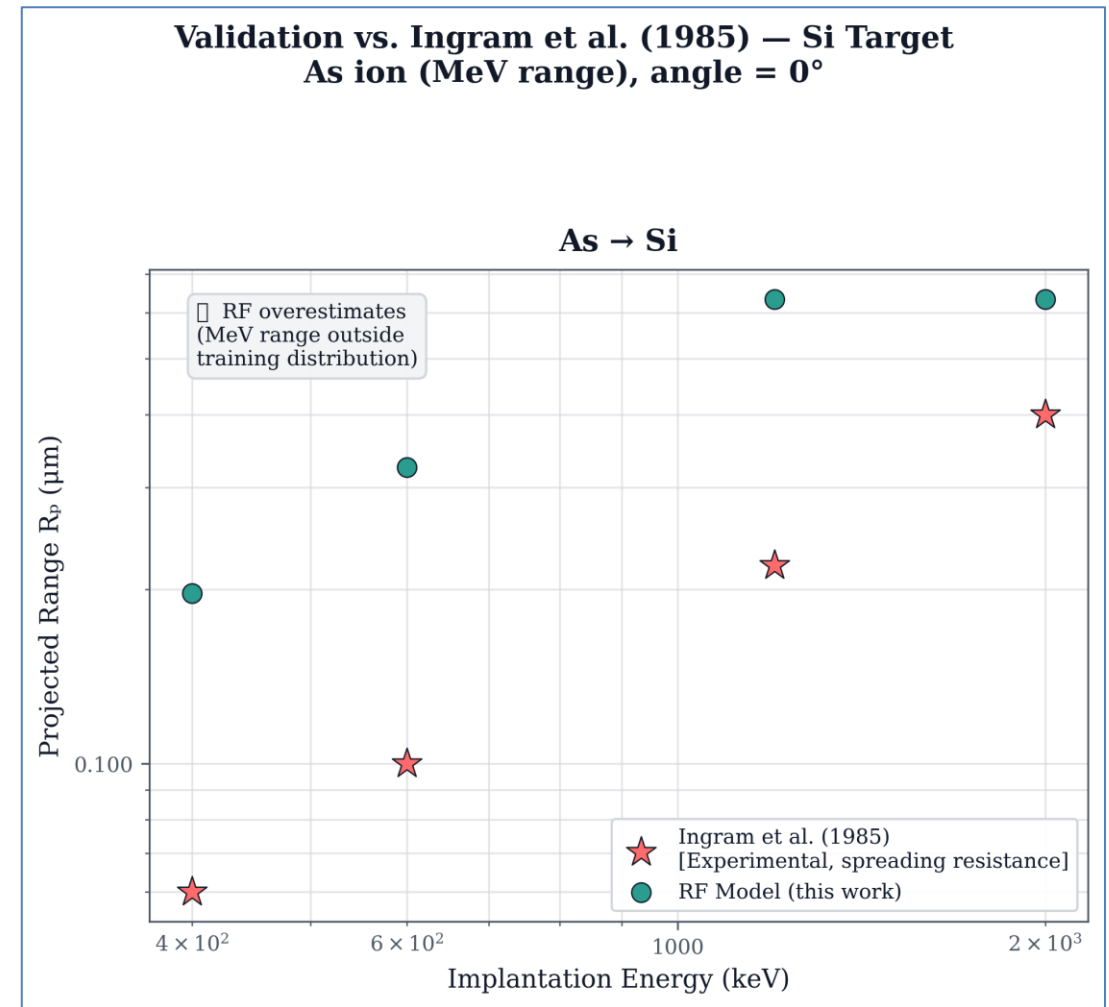
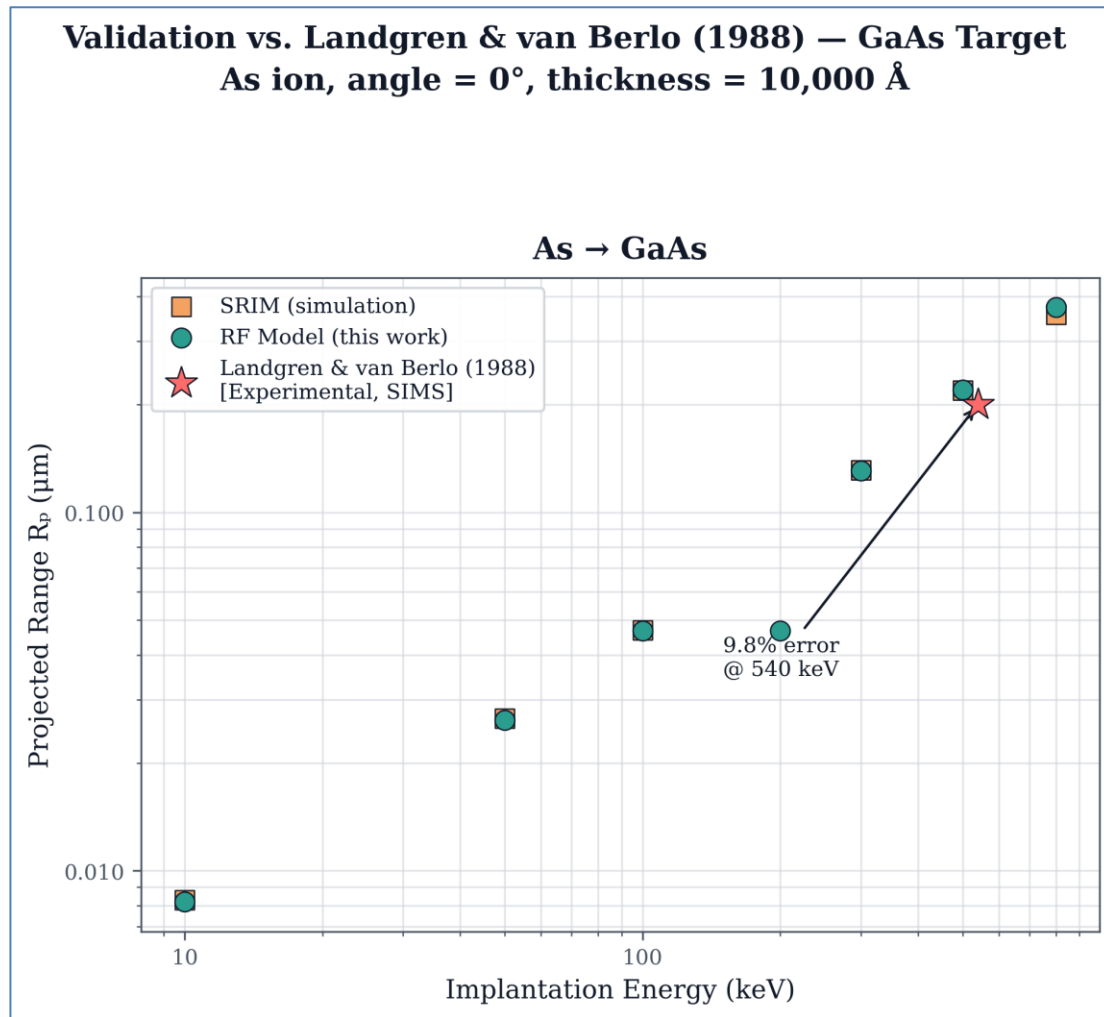


Fig. 24: Experimental SIMS validation with Model output for GaAs and Si substrate

Quantitative Comparison — RF Model vs. Experimental R_p

Table 10. Quantitative comparison with experimental data and model output

Sub.	Ion	E (keV)	Exp. R_p (Å)	SRIM R_p (Å)	RF R_p (Å)	SRIM Err(%)	RF Err(%)
SiC	B	10	293	273	270	6.7	7.9
SiC	B	100	1686	2019	2017	19.7	19.6
SiC	B	500	5729	7202	7249	25.7	26.5
SiC	P	50	411	468	467	13.7	13.6
SiC	P	100	806	904	903	12.2	12.0
SiC	P	300	2339	2507	2512	7.2	7.4
SiC	P	500	3840	4005	4035	4.3	5.1
SiC	P	800	6058	5937	6067	2.0	0.2
SiC	P	1000	7521	7055	7002	6.2	6.9
GaAs	As	540	2000	2176	2196	8.8	9.8
AVERAGE			—	—	—	10.6%	10.9%

Highlights

0.2%

P at 800keV
best match

9.8%

A at 540keV
GaAs

RF≈SRIM

Model learned
SRIM physics

⚠ B: 20–29%

SRIM also same
(known limit)



Annealing Effect

High temperature annealing is a traditional process used to repair the lattice damaged by high-energy ion implantation and activate implanted ions.

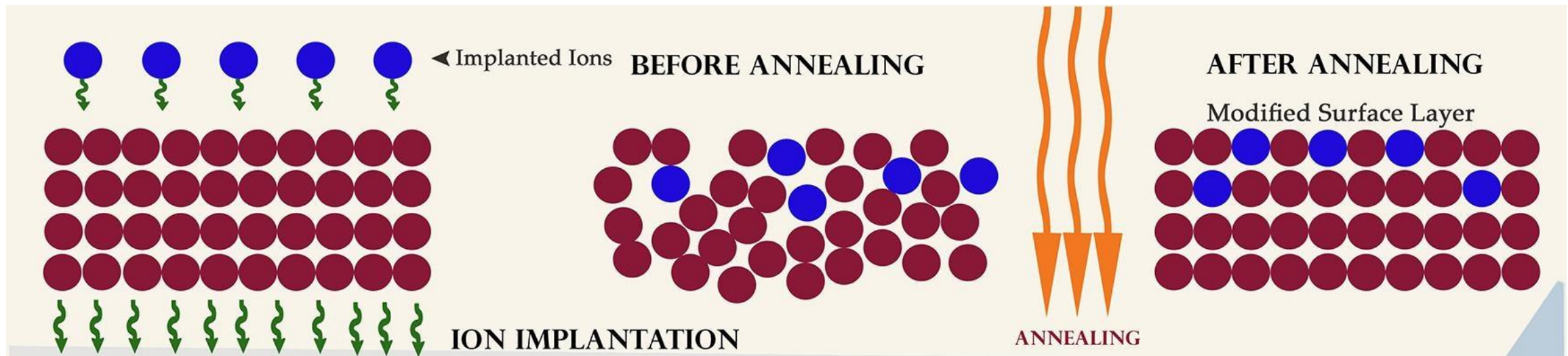
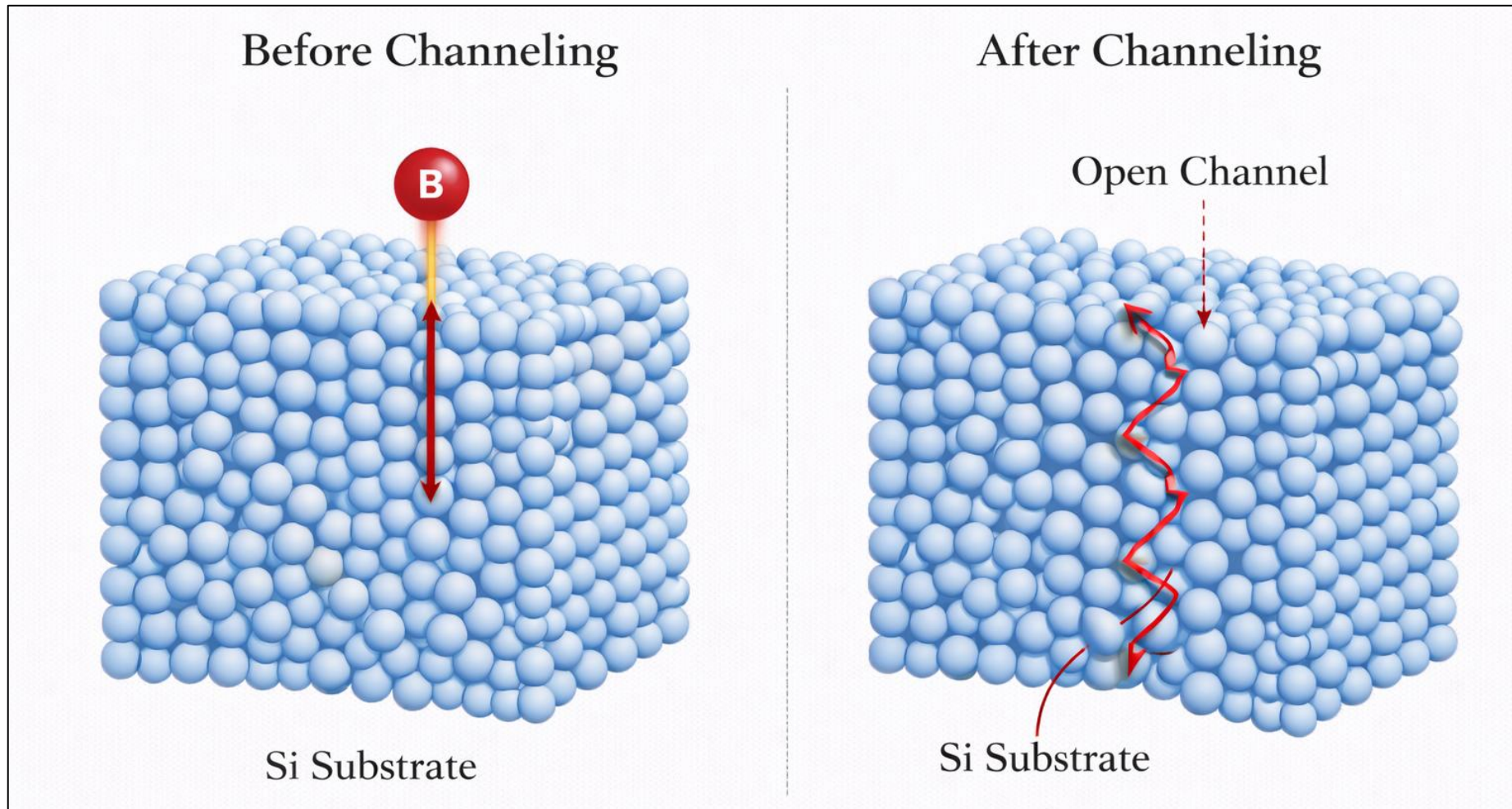


Fig 25. Effect of annealing in ion implantation process.

<https://www.sciencedirect.com/science/article/abs/pii/S0167577X21020541>

Channeling Effect



This figure compares **boron (B) ion implantation in a silicon (Si) substrate** before and after channeling. Before channeling, the ion undergoes frequent collisions and follows a relatively straight, shallow path, whereas after channeling it travels deeper along open crystallographic channels with reduced nuclear scattering.

Fig 26. Channeling effect in ion implantation process.

Substrate and Ion Wise Analysis

Table 11: Gatekeeper Classifier by Substrate (10-Fold CV)

Substrate	Accuracy	Precision	Recall	F1
Silicon (Si)	0.922	0.936	0.964	0.950
SiC	0.934	0.948	0.960	0.954
GaAs	0.927	0.952	0.960	0.956

Table 12: Regression Performance by Substrate

Substrate	Samples	Avg MAE (Å)	Avg RMSE (Å)	Avg R ²
GaAs	1523	477.06	1356.90	0.912
Si	1447	413.79	1136.79	0.907
SiC	1483	312.47	866.72	0.890

Table 13: Ion-Specific Gatekeeper Classification (10-Fold CV)

Ion	Mass Type	Accuracy	Recall (Stopped)	F1	Avg R ² (Reg.)	Avg RMSE (Å)	Physical Challenge
(As)	Heavy (Z=33)	0.987	0.997	0.992	0.941	747	Dense cascades, easy to learn
(Ar)	Medium (Z=18)	0.981	0.995	0.988	0.918	968	Inert / no bonding
(P)	Medium (Z=15)	0.976	0.994	0.985	0.904	995	Mixed stopping regime
(Mg)	Light (Z=12)	0.954	0.981	0.967	0.882	1268	Moderate straggle
Boron (B)	Light (Z=5)	0.939	0.991	0.965	0.867	1546	Sparse damage, high straggle



Inverse Model Evaluation

Table 14: Performance comparison of different machine learning models for inverse prediction system

Model Name	Accuracy (R ² Score)	Global RMSE (keV)	Verdict
Linear Regression	0.18	938.9	Failed. Cannot capture the non-linear relationship between Range and Energy.
KNN (Weighted)	0.95	225.9	Good. Performs well but struggles slightly with interpolation in sparse data regions.
Random Forest	0.96	205.6	Best. Demonstrates the highest accuracy and robustness across all substrates.

Table 15: Error breakdown by energy regime (Random Forest)

Energy Range	RMSE (Error)	Interpretation
Global (All Data)	205.6 keV	Average error is higher due to variance in the MeV range.
Low Energy (< 200 keV)	4.89 keV	High Precision. Excellent performance for standard source/drain implantation.
High Energy (≥ 200 keV)	332.9 keV	Acceptable variance for deep well formation where strict precision is less critical.



Applications

Semiconductor Manufacturing:

Ion implantation is crucial for doping semiconductors, controlling the threshold voltage of transistors, and forming n+ and p+ regions. It's used to create well formations, drain engineering, and to break up native oxides.

Metal Finishing:

Ion implantation improves the wear resistance, corrosion resistance, and fatigue strength of metal parts. This is particularly important in industries like **aerospace, automotive, and precision engineering.**

Materials Science:

It's used to study and *modify the properties* of a wide range of materials, including polymers, ceramics, and even diamond.

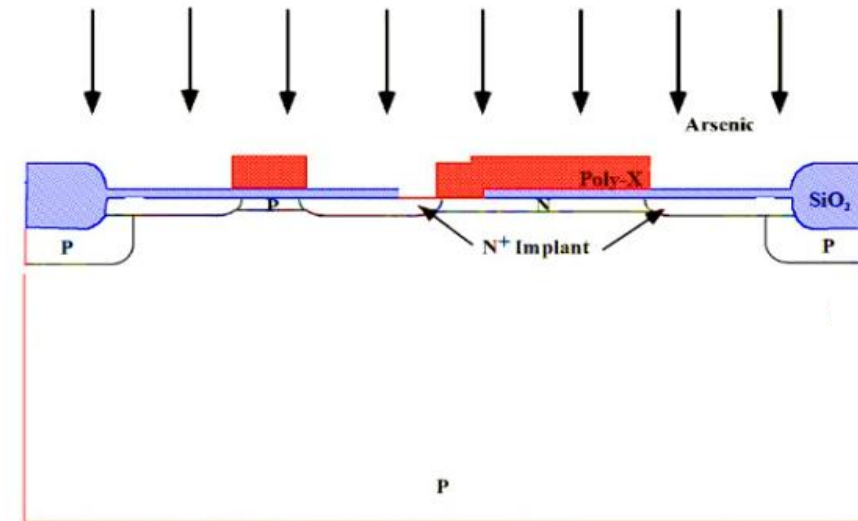


Fig. 27. N+ Implantation in MOSFET.

Experimental Setup

A new ion implanter has been in operation at **The Institutes of the Helmholtz-Zentrum Dresden-Rossendorf** since 2016. The facility with four experimental stations increases the attractiveness of the Ion Beam Center for research and industry as a supraregional center for ion beam applications.



Fig. 28. Ion Implantation setup in lab experiment.

<https://www.hzdr.de/db/Cms?pOid=48616&pNid=0>

Test Example for Forward Model

Substrate = Silicon (Si) | Ion = Boron (B) | Substrate Thickness = 10000 Å | Angle = 7°

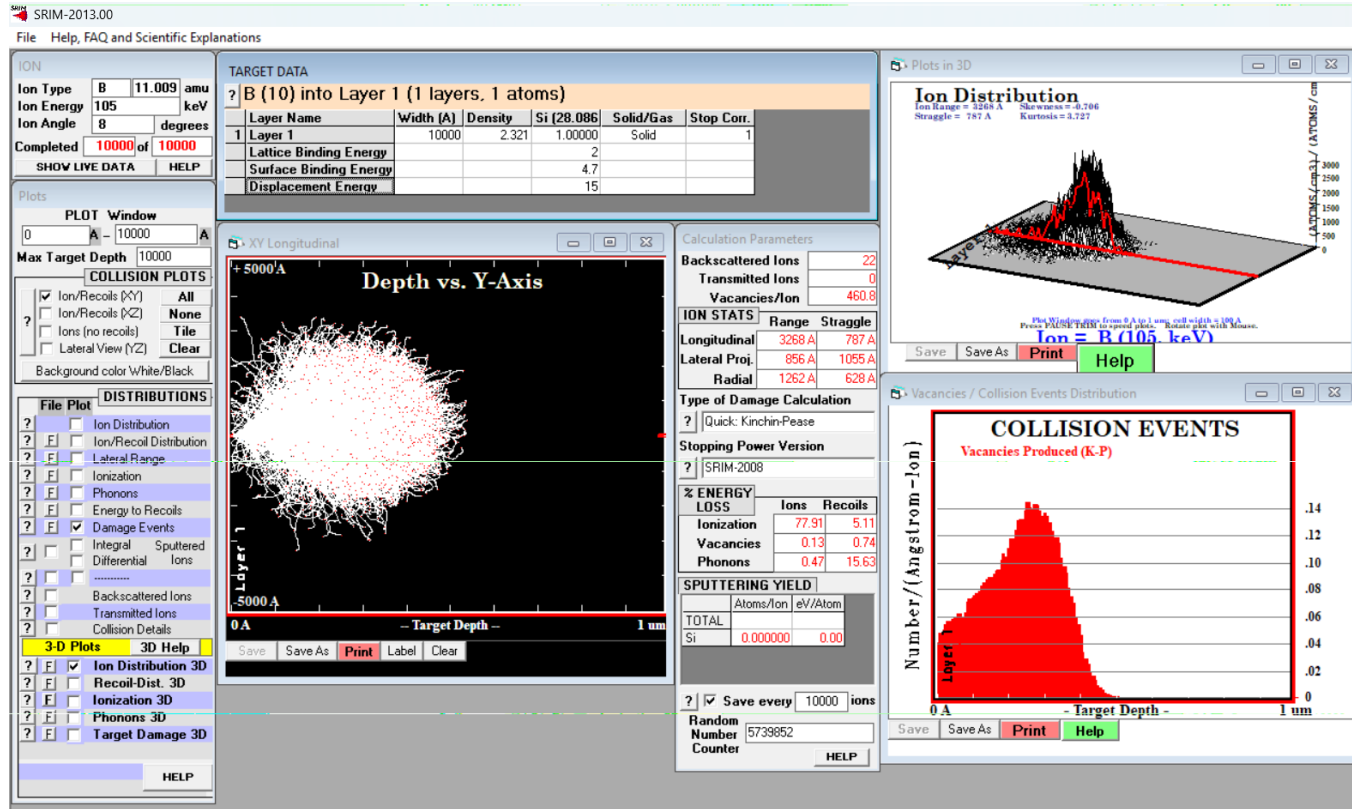


Fig. 29. Simulated interface of SRIM for test example

Table 17: Output comparison with SRIM and Model

Parameter Name	SRIM Simulated	Model Predicted	Absolute Error	% Error
Projected Range	3268 Å	3112.1 Å	155.90 Å	4.77%
Projected Range Straggle	787 Å	770.70 Å	16.30 Å	2.07%
Lateral Range	856 Å	797.03 Å	58.97 Å	6.89%
Lateral Straggle	1055 Å	999.86 Å	55.14 Å	5.23%
Radial Range	1262 Å	1206.8 Å	55.18 Å	4.37%
Radial Straggle	628 Å	605.62 Å	22.38 Å	3.56%
Vacancies per Ion	460.8	489.43	28.63	6.21%
Backscattered Ions	22	25.53	3.53	13.8%
Transmitted Ions	0	0	0	0%

Test Example for Inverse Model

Scenario: A process engineer needs to implant Boron into Silicon with a target junction depth of $R_p^* = 500 \text{ \AA}$. What implant recipe should they use?

1 Input to Inverse Solver:

M = Silicon (Si) | $R_p^* = 500 \text{ \AA}$ | T (max thickness) = 2000 $\text{\AA}$ | S = enumerate all ions

2 Solver Enumerates (S, ?) Combinations & Predicts Energy via RF:

Ion (S)	Angle (?)	Predicted Energy (keV)	Feasible?	Notes
Boron (B)	0°	~45 keV	☑ Yes	Standard implant industry common
Boron (B)	7°	~47 keV	☑ Yes	7° tilt reduces channeling (recommended)
Boron (B)	30°	~60 keV	☑ Yes	Higher E needed to compensate tilt
Phosphorus (P)	0°	~95 keV	☑ Yes	Heavier ion needs more energy for same depth
Arsenic (As)	0°	~250 keV	☑ Yes	Heaviest ion highest energy required
Magnesium (Mg)	89.9°	> 10.000 keV	✗ Rejected	Outside feasible energy range filtered out

